



Adaptive thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds[☆]

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ABSTRACT

Early diagnosis of respiratory diseases through sound analysis is often challenged by noise from various internal and external sources. Discrete wavelet transform (DWT)-based denoising has shown significant potential for noise suppression, but its effectiveness heavily depends on the selection of thresholding parameters. To address this, we introduce a novel adaptive thresholding approach of the DWT coefficients using a UNet model for respiratory sound denoising. The UNet maintains spatial localization while integrating contextual information across different levels, allowing precise and context-aware thresholding that effectively suppresses noise while preserving signal components. Combining the multi-resolution capabilities of DWT with UNet's multi-scale feature extraction, the method demonstrates its robustness and efficacy on two public datasets with various noise types and levels, outperforming traditional methods. The method achieved a 2.0 dB higher SNR than the second-best result, leading to a 3.45% improvement in sound event classification accuracy under real-life noise conditions. These results highlight its potential for deployment in rural and remote areas. The code for the model is available at https://github.com/subhasikta/DWT-UNet_denoising_model.

1. Introduction

Respiratory diseases are progressive conditions affecting the lungs and various parts of the respiratory system, significantly impacting the quality of human life. Respiratory diseases, such as chronic obstructive pulmonary disease (COPD), tuberculosis, lung cancer, asthma, and acute lower respiratory tract infections (LRTI), occur due to multiple factors such as infection, smoking, exposure to pollutants, and other harmful substances. According to the world health organization (WHO), respiratory diseases are one of the most common causes of mortality in the world, accounting for more than 3 million deaths every year worldwide [1]. Therefore, developing accurate and efficient techniques to detect respiratory disorders is crucial to curtail their burden. Several clinical methods are available for diagnosing respiratory disorders, such as chest X-rays, pulmonary function tests (PFTs), and computed tomographic (CT) scans, which offer crucial insights into the chest's anatomy [2]. However, these methods are not always readily available due to the high cost of equipment, especially in rural areas, and they also involve the risk of harmful radiographic effects.

Auscultation is another approach where healthcare professionals hear respiratory sounds by placing a stethoscope on the patient's chest

or back during inhalation and exhalation. The stethoscope is a non-invasive, low-cost, easy-to-use, and lightweight equipment suitable for rural and remote areas [3]. Respiratory sounds, also known as lung sounds, are produced by the airflow in the lungs during respiration and provide critical indicators of respiratory health [4]. Respiratory sounds are broadly classified into normal and abnormal sounds. Normal lung sounds are divided into four categories: tracheal, bronchial, bronchovesicular, and vesicular. On the other hand, abnormal sounds, including wheeze, stridor, rhonchi, fine crackle, coarse crackle, and pleural friction rub, are related to respiratory disorders such as narrowing of airways, obstruction, or fluid in the airway [5,6].

Accurate assessment of respiratory sounds faces challenges due to potential contamination from external noises such as environmental noise, stethoscope artifacts, or body movement [7] and internal noise due to heart functionality [8]. A robust to-noise system will enable users to monitor respiratory health frequently or in the long term in real-life scenarios, leading to early diagnosis of respiratory diseases. Therefore, it is crucial to suppress these unwanted sounds to enhance the signal quality.

Removing noise from respiratory sound signals is challenging due to their chaotic and dynamic nature, further complicated by the overlap

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of heart sounds in time and frequency domains [9]. Various methods have been explored to address these issues. However, these methodologies retain inherent limitations. For example, filter-based methods [7,10–12] often struggle with non-stationary noise and overlapping frequency ranges, leading to a trade-off between noise reduction and signal distortion. To exploit the time-frequency localization, Lee et al. [12] developed a wavelet-based empirical Wiener hybrid method, while Gradolewski et al. [13] proposed wavelet transform-based methods to figure out the non-stationary characteristics of heart and lung sounds. Despite promising results, time-frequency localization approaches face limitations in selecting optimal thresholding parameters [9,14]. Methods based on empirical mode decomposition (EMD) suffer from mode mixing, parameter sensitivity, and computational complexity [4,15]. Also, these methods need domain-specific tuning of the filtering parameters, which is a non-intuitive task.

Recent studies increasingly leverage deep learning-based approaches to automate noise removal. For instance, Hsieh et al. [16] proposed WaveCRN, a time-domain method that integrates 1D convolution neural network (CNN) with bidirectional simple recurrent units (SRUs) to estimate clean speech directly from the raw signal by leveraging its inherent information. Sun et al. [17] separated heart and lung sounds using non-negative matrix factorization (NMF) by extracting features using a DAE, then utilizing variational mode decomposition (VMD) to denoise the lung sound signal. Hu et al. [18] employed a fully-convolutional time-domain audio separation network (Conv-TasNet) for end-to-end single-channel speech separation, which directly operates on waveforms using a linear encoder–decoder framework and a temporal convolutional network (TCN) to model long-term dependencies. Yang et al. [19] introduced a two-stage framework, where an adaptive noise cancellation (ANC) module performs coarse denoising of auscultation signals, followed by a cascaded dual-channel interactive noise cancellation network (DINC-Net) that eliminates residual noise further, inspired by [18]. Despite these advancements, many existing methods fail to fully exploit time-frequency localization for noise separation or rely on thresholding with predefined functions.

Leveraging the time-frequency representation, Bae et al. [20] proposed WavResNet, a residual CNN framework that estimates noise sub-bands in the wavelet domain. These estimated noise sub-bands are subtracted from the input sub-bands, followed by an inverse DWT to reconstruct the denoised image. Luo et al. [21] proposed the deep complex convolution recurrent network (DCCRN) to process the complex-valued spectrogram for speech quality enhancement. Tsai et al. [22] processed the spectrogram using a periodically encoded deep autoencoder (DAE) for heart-lung sound separation. Pouyani et al. [9] utilized an artificial neural network (ANN) to filter DWT coefficients adaptively, but this approach delegates the reconstruction phase to the ANN, adding complexity and limiting denoising efficacy. Additionally, these methods overlook abnormal cases and lack thorough validation across diverse noise environments. Most of these models also have limitations, as they are not validated on various noise levels and are ineffective in denoising abnormal respiratory sounds since they do not consider these cases.

Recently, the UNet architecture has been explored for denoising due to its multi-scale feature extraction capability. Yang et al. [23] proposed a complex-valued U-net (CUNet) with a DAE structure, while Wang et al. [24] process the short-time Fourier transform (STFT) images using the sparse non-negative matrix factorization (SNMF) and a multi-channel UNet to separate heart and lung sounds. However, the application of UNet models integrated with time-frequency representations remains relatively under-explored. Nakamura et al. [25] proposed W-UNet, an architecture that incorporates the DWT into the UNet framework to replace traditional pooling and sampling operations. This integration mitigates aliasing effects while preserving feature map information. The W-UNet concept has since been adopted in various domains, including signal denoising [26], image denoising [27,28], and image segmentation [29–31]. Despite their success, these methods

do not fully exploit the multi-resolution capabilities of the DWT for effective feature representation and denoising purposes.

Unlike prior models, the proposed approach introduces a novel adaptive thresholding technique for the DWT coefficients of respiratory sound signals using a UNet-based framework, thereby eliminating the need for manual selection of optimal thresholding parameters. To the best of the authors' knowledge, this is the first study to employ a UNet architecture for adaptive thresholding of wavelet coefficients. UNet is well-suited for thresholding DWT coefficients due to its multi-scale feature extraction capability, which complements the DWT's decomposition of signals into various frequency bands. The UNet architecture maintains spatial localization while integrating contextual information across different levels, allowing precise and contextually informed thresholding decisions that effectively suppress noise while preserving essential signal components. By combining the multi-resolution capabilities of DWT with the multi-scale feature extraction strengths of UNet, our method offers a more robust and effective solution for real-world respiratory sound analysis.

The main contributions of this work are summarized as follows:

- (1) The work introduces an adaptive thresholding approach for DWT coefficients of respiratory sound signals using a UNet-based framework, thus alleviating the need for optimal filtering parameter selection.
- (2) Experiments were conducted on normal and abnormal signals contaminated with various synthetic and real-life noises. Additionally, the impact of the denoising on the classification is tested.
- (3) Thresholding DWT coefficients of each level using a UNet achieves superior denoising results than existing methods due to the multi-scale feature extraction capability.

2. Methodology

The block diagram of the proposed approach is shown in Fig. 1. First, the respiratory sound signals are pre-processed to get the clean signals, and then various synthetic and real-life noises are added to these sounds. In the denoising step, a band-pass filter is first applied to remove out-of-band noise, and then the signal is decomposed using the DWT up to four levels. Subsequently, the level-1 detail coefficients (D_1) and approximation coefficients (A) are set to zero, while the noisy detail coefficients from levels D_2 , D_3 , and D_4 are passed through separate UNet models. This multi-branch UNet architecture enables the network to learn scale-specific noise characteristics unique to different frequency bands. After thresholding is performed by the UNet model, the inverse DWT is applied to reconstruct the denoised signal in the time domain. To evaluate the effect of denoising on downstream classification, a Decision Tree (DT) classifier model [32] is employed to categorize the sound signals into respective classes. Detailed descriptions of each step are provided in the following subsections.

2.1. Dataset description

This study uses the following two popular open-access datasets.

(i) *SJTU Pediatric Respiratory Sound 2023 (SPRS-23)* [33]: It comprises 2820 sound records acquired from 346 subjects at four different posterior locations with a sampling frequency of 8 kHz. The dataset includes label annotations at two levels: record level and event level. At the record level, sound records are classified as normal (1842), continuous adventitious sound (CAS) (261), discontinuous adventitious sound (DAS) (381), CAS & DAS (116), and poor quality (220). The duration of these sound records ranges from 3 to 15.36 s. Each sound record contains multiple sound events with durations ranging from 137 ms to 7.15 s. There are a total of 9780 sound events, including normal (7617), rhonchi (134), wheeze (646), stridor (47), coarse crackle (83), fine crackle (1221), and wheeze & crackle (32).

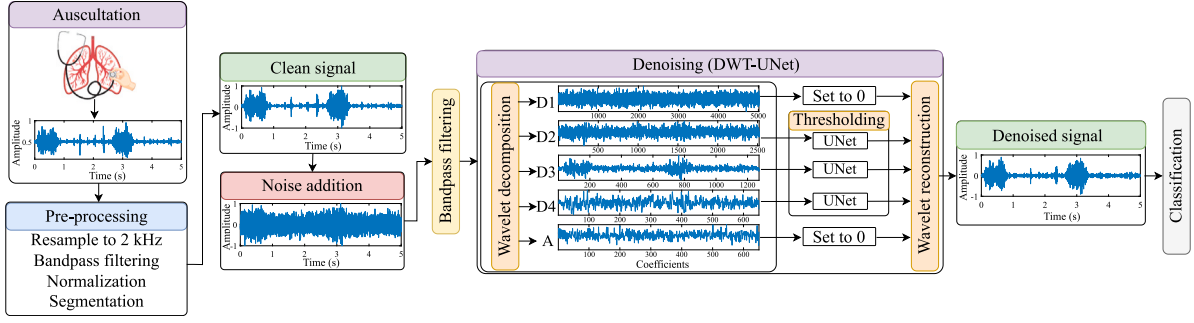


Fig. 1. Block diagram of the proposed approach.

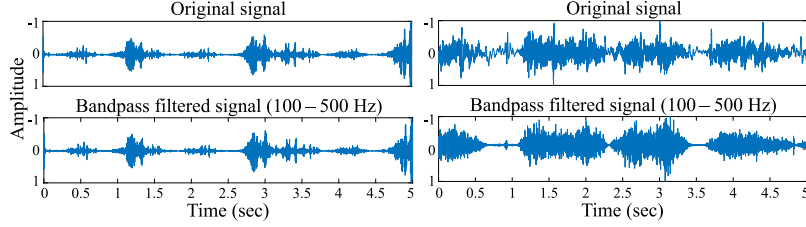


Fig. 2. Original signal and bandpass signal (left) SPRS-23 and (right) ICBHI-17.

(ii) *International Conference on Biomedical and Health Informatics 2017 (ICBHI-17)* [34]: It comprises 920 sound records acquired from 126 subjects at both anterior and posterior locations with a sampling frequency ranging from 4 kHz to 44.1 kHz. These sound records vary in duration, ranging from 7 to 90 s. Each sound record is segmented into multiple sound events with durations ranging from 542 ms to 16.16 s. There are a total of 6898 sound events, including normal (3642), crackle (1864), wheeze (886), and wheeze & crackle (506).

2.2. Data pre-processing

The datasets used for the study have varying sampling frequencies ranging from 4 kHz to 44.1 kHz. Since the frequency of lung sounds typically falls below 1 kHz [14], the signals were down-sampled to a 2 kHz sampling rate. As illustrated in Fig. 2, the frequency range of lung sounds lies in the 100–500 Hz frequency band in both datasets, and out of this band, there is high noise contamination. Therefore, a 6th-order Butterworth filter with a 100–500 Hz pass-band is applied to avoid low-frequency and high-frequency noise. Based on the observation of spike noise in the initial portion of some recordings, the first 1 s of each sample in the SPRS-23 dataset was discarded during pre-processing. Respiratory sound records varied in duration; therefore, recordings were segmented into 5 s segments to maintain the uniformity of length, and the last segment shorter than 5 s, if any, is padded with zeros to make it a 5 s segment. Further, min–max normalization is used to rescale the amplitude of signals on the same scale, removing differences caused by varying devices and geographical locations.

2.3. Noise addition

After the pre-processing step, synthetic and real-life noise is added to the signals. Synthetic noise was introduced using three noise models: Additive white Gaussian noise (AWGN), pink noise, and red noise. The noise was added to produce noisy signals with distinct signal-to-noise ratio (SNR) values of −5 dB, 0 dB, and 5 dB. White noise is the best simulation of environmental noise when recording sounds, whereas pink and red noises mimic the characteristics of real-world scenarios [35].

The real-world noise is also added to the signals to analyze the performance of the proposed method in real-life scenarios. For this,

a publicly available ARCA23K dataset [36] consisting of real-world noise sound is used. There are 23,727 audio samples labeled with 70 different real-world sound classes and sampled at 44.1 kHz. Among these classes, only a subset of classes expected to occur in daily-life environments are considered, such as clapping, coughing, female speech, male speech, footsteps, and child speech. These sound signals were resampled to 2 kHz to match the sampling rate of the respiratory signals and normalized to the range [−1, 1] before adding them to the respiratory sound signals.

2.4. DWT-UNet denoising model

The proposed denoising method first applies a 6th-order Butterworth band-pass filter with a cutoff frequency of 100 to 500 Hz to remove out-of-band frequency components. Then, the filtered signal is decomposed using DWT up to four levels following Mallat's sub-band coding algorithm [37], as shown in Fig. 3. The DWT coefficients are then thresholded adaptively using the UNet models. Finally, the thresholded coefficients are used to reconstruct the denoised signal. A detailed description of these steps is provided below.

DWT decomposition: The signal is decomposed into approximation and detailed coefficients using DWT up to four levels following Mallat's sub-band coding algorithm [37], as shown in Fig. 3. The DWT decomposes the signal into multiple frequency sub-bands, allowing a multi-resolution analysis of the signal in the time-frequency domain. As per Mallat's algorithm, the first level of approximation and detail coefficients are obtained by applying analysis filters consisting of a low-pass filter (h) and a high-pass filter (g). These approximation and detail coefficient outputs are down-sampled by a factor of two. Then, both analysis filters are applied to the decimated approximation coefficients to further obtain the coefficients at the next level. The approximation and detail coefficients are calculated as follows:

$$y_{\text{high}}[k] = \sum_n x[n] \cdot g[-n + 2k] \quad (1)$$

$$y_{\text{low}}[k] = \sum_n x[n] \cdot h[-n + 2k] \quad (2)$$

Where $x[n]$ represents the original signal, and $y_{\text{high}}[k]$ and $y_{\text{low}}[k]$ represent the outputs of the high and low-pass filters, respectively. $g[-n+2k]$ and $h[-n+2k]$ denote the high and low-pass filter coefficients, respectively, being shifted by n and scaled by $2k$.

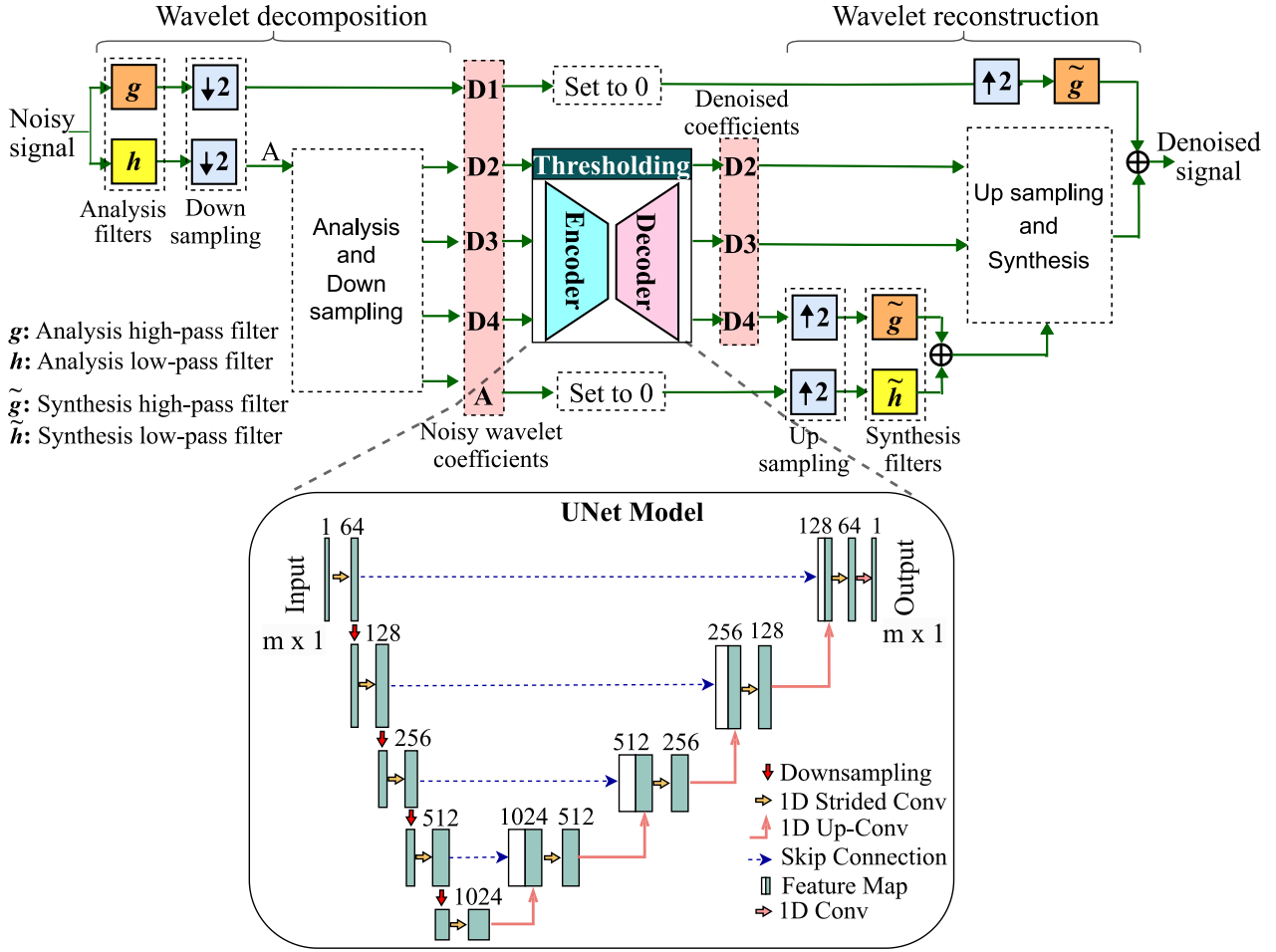


Fig. 3. Framework of the proposed DWT-UNet model.

This study used the ‘db14’ (Daubechies with fourteen vanishing moments) wavelet as the mother wavelet to decompose the signals up to four levels. Decomposing a signal via DWT splits its frequency range into halves at each level. For a 2 kHz sampling frequency, the detailed coefficients for levels 1 through 4 correspond to frequency ranges of approximately 500–1000 Hz, 250–500 Hz, 125–250 Hz, and 62–125 Hz, respectively. The approximation coefficients (A) cover the frequency range of 0–62 Hz, and the detail coefficients at level one (D_1) capture the 500–1000 Hz frequency range, where the respiratory contents are minimal and the noise dominant. Therefore, these coefficients are set to zero, and only detailed levels D_2 , D_3 , and D_4 are thresholded using the UNet model. A 5 s signal with 10,000 samples decomposed up to four levels using Mallat’s sub-band coding yields coefficient lengths of 5013 (D_1), 2520 (D_2), 1273 (D_3), 650 (D_4), and 650 (A). The slightly more-than-halved sizes at each level are expected due to convolution and boundary effects during filtering and down-sampling, aligning with Mallat’s decomposition behavior [37].

UNet for DWT coefficients thresholding: This study proposes to use a UNet model for adaptive thresholding of the DWT coefficients for the first time as of the author’s knowledge. The detailed levels D_2 , D_3 , and D_4 noisy coefficients are fed to separate UNet models, allowing them to focus on distinct frequency components and different noise characteristics at each level. The UNet model, proposed by Ronneberger et al. [38] in 2015, comprises the encoder and decoder layers in a “U” shape. The encoder layer stepwise down-samples the feature map of the input image to create a compact latent representation, while the decoder layers stepwise up-sample the feature maps to produce outputs of the same size as the input. The last layer, called the bottleneck, acts as a bridge between the encoder and decoder. In addition, skip

connections connect the encoder layer to the decoder layer, helping retain the high-resolution contextual features.

The UNet takes DWT coefficients of size $m \times 1$ as input and produces thresholded (denoised) coefficients of the same size as output, where m is the number of wavelet coefficients at a particular detailed level. Each layer in the encoder employs a 1D convolutional layer with a kernel size of 3, a stride of 2, and zero padding. In this convolution approach, the stride of 2 is used to reduce the size of the feature map instead of a separate max-pooling layer to retain finer details and ensure smoother gradient flow. A dropout layer is also applied by giving a dropout rate of 50% to prevent overfitting, followed by the LeakyReLU activation function.

The decoder up-samples the coefficients using 1D transposed convolutions and merges them with the encoder feature maps via skip connections. Later, a final convolution layer is applied to combine the features from the decoder and the skip connection to get the final output. The output layer utilizes a linear activation function to produce outputs with continuous values. The UNet model is fed with the noisy signal’s DWT coefficients and trained to deliver the DWT coefficients of the clean signal. Thus, the UNet model learns to threshold the DWT coefficients by suppressing noisy components and preserving the useful signal components. The model is trained to reduce the mean squared error (MSE) loss between the predicted and clean signal DWT coefficients, as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (3)$$

where $\hat{y}_i$ is the predicted DWT coefficients and y_i is the DWT coefficients of the clean signal.

Instead of thresholding the DWT coefficients with a fixed calculated threshold value as in the case of traditional DWT-based denoising methods, the proposed method offers a framework for adaptive thresholding of the DWT coefficients to effectively denoise respiratory sounds.

Reconstruction of the signal: The thresholded wavelet coefficients produced by the UNet model undergo an inverse DWT operation to reconstruct the denoised time-domain signal. As previously mentioned, the D_1 and A coefficients are set to zero, and thresholded D_2 , D_3 , and D_4 coefficients are used to reconstruct the denoised time-domain signal. Finally, the denoised 5 s segments are concatenated to construct the full-length signal. The padding will be removed from the last segment, if any, to form the denoised signal with the same length as the input signal.

2.5. Classification of respiratory sound signals

The denoised signals are classified at the sound event level to demonstrate the impact of the proposed denoising method on event classification. The classification is performed using a DT classifier proposed in [32]. For classification tasks, thirteen mel-frequency cepstral coefficients (MFCCs) are extracted from the denoised signals, followed by the computation of the mean and standard deviation of these coefficients. The features were then normalized and used to train the DT classifier, which was initialized with a minimum leaf sample size of 25.

3. Experimental results

Comprehensive experiments are conducted on two datasets with different types of synthetic and real-life noise, for which the experimental setup and results are presented in the following subsections.

3.1. Experimental setup

The experimental setup used for the study is as follows.

Data formation: This study utilizes two publicly available respiratory sound datasets: SPRS-23 and ICBHI-17. The SPRS-23 dataset is pre-organized into two folders: one for training, containing 1949 samples, and another for testing, containing 871 samples. The ICBHI-17 dataset was divided into training and testing sets at an 80:20 ratio, resulting in 736 and 184 samples, respectively. Signals were then segmented into 5 s segments, with shorter recordings padded with zeros, resulting in 4414 training and 2458 testing segments in the SPRS-23 dataset and 3195 training and 819 testing segments in the ICBHI-17 dataset. These segments were contaminated with AWGN, pink, and red noise at three SNR levels (−5 dB, 0 dB, and 5 dB) and real-life noise. Therefore, this resulted in 44,140 ($4414 \times (3 \text{ types} \times 3 \text{ levels})$) synthetic + 1 real-life noise) and 31,950 ($3195 \times (3 \text{ types} \times 3 \text{ levels})$) synthetic + 1 real-life noise) noisy samples for the SPRS-23 and ICBHI-17 datasets, respectively. These extensive sets of noisy signals were used to train the UNet model, without inclusion of clean signals. The trained model was subsequently evaluated on test sets, each contaminated separately with a specific type and level of noise. The average performance metric was computed across 2458 test samples for SPRS-23 and 819 test samples for ICBHI-17.

System specification and hyperparameters: The proposed model was implemented using TensorFlow™ and Keras™ libraries on a system equipped with an Intel Xeon® Silver 4216 processor, 128 GB RAM, and an Nvidia™ RTX A000 GPU (16 GB VRAM) for accelerated computation. The signal processing, noise addition, DWT decomposition, and reconstruction are performed using the Wavelet Toolbox™ [39] of MATLAB™ R2023b software.

Table 1 describes the parameters of the proposed denoise model used in this study. The UNet models for D_2 , D_3 , and D_4 levels are trained for 3000, 1500, and 1000 epochs, respectively. As D_2 has more coefficients and is complex, and needs more training for efficient feature extraction,

Table 1

Hyperparameter configurations for the DWT-UNet.

	D_2	D_3	D_4
Input Size	2520×1	1273×1	650×1
Output Size	2520×1	1273×1	650×1
Epochs	3000	1500	1000
Learning Rate	0.0001		
Optimizer	Adam		
Batch Size	64		
Loss	MSE		

while D_3 and D_4 have fewer coefficients, they require fewer epochs to avoid overfitting and to ensure efficient training.

Performance evaluation metrics: Denoising performance is evaluated using three quantitative parameters: SNR [40], Fit (*Fit*) coefficients [41], and Correlation (*Corr*) coefficients [9]. SNR measures the ratio of the strength of the signal to that of background noise, with higher values indicating a lower level of noise contamination relative to the signal's strength. *Fit* quantifies how closely the denoised signal matches the original clean signal, with a higher value indicating a better match. *Corr* assesses the linear relationship between the denoised and original signals. A value of *Corr* close to 1 indicates a strong positive linear relationship, which implies that the denoised signal closely follows the clean signal pattern. The following are the mathematical expressions for these metrics.

$$SNR(y_{cs}, y_{ds}) = 10 \log_{10} \left(\frac{\sum_{i=1}^{len} y_{cs}(i)^2}{\sum_{i=1}^{len} (y_{ds}(i) - y_{cs}(i))^2} \right) \quad (4)$$

$$Fit(y_{cs}, y_{ds}) = 100 \times \left(1 - \frac{\sum_{i=1}^{len} (y_{ds}(i) - y_{cs}(i))^2}{\sum_{i=1}^{len} (y_{cs}(i) - \mu_{y_{cs}})^2} \right) \quad (5)$$

$$Corr(y_{cs}, y_{ds}) = \frac{\text{cov}(y_{cs}, y_{ds})}{\sigma_{y_{cs}} \sigma_{y_{ds}}} = \frac{E[(y_{cs} - \mu_{y_{cs}})(y_{ds} - \mu_{y_{ds}})]}{\sigma_{y_{cs}} \sigma_{y_{ds}}} \quad (6)$$

where len is the number of samples in a signal, y_{cs} represents the original clean signal, and y_{ds} represents the obtained denoised signal. $\mu_{y_{cs}}$ and $\sigma_{y_{cs}}$ indicate the mean and standard deviation of the clean signal, and $\mu_{y_{ds}}$ and $\sigma_{y_{ds}}$ are the mean and standard deviation of the denoised signal.

The SNR provides the overall quality of the signal, while *Fit* and *Corr* parameters indicate the element-wise performance of the methods. Compared to *Fit*, *Corr* also considers the $\sigma_{y_{ds}}$ in the denominator, so if the denoised signal has a lower deviation, the *Corr* will be higher. i.e. If the *Corr* is higher than the *Fit*, the denoised signal has a lower deviation than the original signal, which is undesirable.

3.2. Comparison methods

The comparison is carried out with three different state-of-the-art denoising methods in various noise case scenarios described below.

DWT-SURE [13]: It is based on the DWT-based denoising method, which decomposes the noisy signal into five different scales utilizing the Coiflet-5 (coif5) as the mother wavelet and employs the Stein's unbiased risk estimate (SURE) estimation method with a soft thresholding function.

DCCRN [21]: This method processes the STFT spectrogram of the noisy signal to produce the denoised spectrogram using a complex encoder-decoder structure coupled with the complex-valued convolution operations. The model also integrates a complex long-short term memory (LSTM) layer between the encoder and decoder layers to capture temporal dependencies. The denoised signal is obtained by applying the inverse STFT to the obtained denoised spectrogram.

WaveCRN [16]: It utilizes CNN to extract locality features of the signal, an simple recurrent unit (SRU) module to model sequential

Table II
Quantitative results of denoising on SPRS-23 dataset.

	AWGN						Pink noise						Red noise						Real-life noise		
SNR before denoise	-5 dB			0 dB			-5 dB			0 dB			-5 dB			0 dB			-		
Performance metrics	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)
DWT-SURE [13]	3.30	51.13	69.77	4.46	57.23	71.14	2.18	43.71	61.10	3.61	52.62	68.22	6.07	70.69	81.33	7.22	75.68	86.36	2.38	50.11	66.05
DCCRN [21]	2.68	43.94	64.02	5.61	62.87	80.98	2.27	49.65	68.54	4.88	55.35	76.30	11.18	86.99	92.39	12.60	87.32	94.80	1.01	36.44	61.97
WaveCRN [16]	5.94	77.33	83.91	7.48	82.27	89.63	5.73	75.51	81.40	6.95	79.06	88.31	11.45	96.17	98.04	13.77	97.32	98.91	5.79	73.44	82.26
DWT-ANN [9]	4.45	64.09	75.52	5.32	70.94	79.31	3.21	59.61	70.83	4.30	68.33	76.01	7.67	80.53	86.62	8.56	83.16	90.11	3.18	54.65	72.18
STFT-UNet [24]	5.01	70.28	79.71	5.96	74.91	81.86	4.61	67.68	75.82	5.24	72.04	80.65	8.91	85.80	91.33	9.03	88.57	93.80	3.91	65.90	79.35
Proposed method	9.09	91.33	95.54	11.46	93.30	96.76	8.93	91.06	95.10	10.23	92.81	96.62	16.95	98.06	99.29	18.98	99.47	99.75	7.84	90.50	94.27
Improvement	+3.1	+14.0	+11.6	+3.9	+11.0	+7.1	+3.2	+15.5	+13.7	+3.2	+13.7	+8.3	+5.5	+1.8	+1.2	+5.2	+2.1	+0.8	+2.0	+17.0	+12.0

dependencies efficiently, and applies a restricted feature masking approach to enhance hidden feature maps, achieving effective lung sound denoising and compressed lung sound restoration.

DWT-ANN [9]: The method first decomposes the noisy signals into eight levels using ‘db14’ mother wavelet. These coefficients are fed into the ANN with two hidden layers, where 35 and 10 neurons are utilized in the first and second hidden layers, respectively. This produces the denoised signal in the time domain and prevents the need for an inverse DWT, improving the performance compared to conventional DWT-based approaches.

STFT-UNet [24]: Similar to DCCRN, this approach also processes the STFT spectrogram image of the noisy signal to obtain the spectrogram of the denoised signal. However, it leverages the UNet model to process the spectrogram image in contrast to the complex hybrid architecture used in DCCRN.

3.3. Experimental results for denoising

The subsection below provides a detailed analysis of the qualitative and quantitative results of the SPRS-23 and ICBHI-17 datasets.

3.3.1. Results on SPRS-23 dataset

The qualitative result are present in Fig. 4, where the noisy signals contaminated with (a) AWGN at -5 dB, (b) Pink noise at -5 dB, (c) Red noise at -5 dB, and (d) Real-life noise and the denoised signal obtained by various methods are depicted. The proposed method demonstrates superior noise reduction in all the noise cases while preserving respiratory contents. In contrast, the traditional DWT-SURE method significantly loses signal information due to the threshold. DWT-ANN methods show improvement but struggle to suppress noise sufficiently, whereas DCCRN could only suppress low-frequency noise, while this method struggles to remove the high-frequency noise. The STFT-UNet also suffers from signal loss, specifically in the presence of real-life noise. The WaveCRN achieves the best results among comparison methods. Notably, the existing methods disrupt the respiratory pattern, whereas the proposed approach successfully denoises signals while maintaining signal patterns across all noise types.

Table II presents the quantitative results of the denoised signals, averaged across test data, where the proposed method achieved the highest SNR, *Fit*, and *Corr* values in all cases of noise at different levels. The proposed method improved the SNR level of the signal from -5 dB to 9.09 dB (AWGN), 8.93 dB (pink noise), and 16.95 dB (red noise). The proposed method generally scores above 90% for *Fit* and *Corr*. Specifically, in the case of real-life noise, the proposed method produces 7.84 dB SNR, 90.50% *Fit*, and 94.27% *Corr*, indicating the method’s efficacy for real-life noise. Notably, the proposed method produces *Corr* values close to *Fit*, while the compared methods exhibit a significant difference between these two parameters in most cases. This indicates that the proposed method produces a denoised signal with a standard deviation close to the original signal. In contrast, the compared methods produce the denoised signal with a lower deviation than the original signal.

Fig. 5 depicts the performance metrics across three noise types evaluated over SNR levels ranging from 5 to -5 dB. The x -axis represents the SNR before denoising, while the y -axis represents the denoised signal's SNR, *Fit*, and *Corr*. The STFT-UNet (black color), DWT-ANN (magenta color), DCCRN (golden-orange color), and DWT-SURE (cyan color) are less effective across noise scenarios. Although the WaveCRN (brown color) performs better across all noise types, it is inferior to the proposed method. The proposed method (green color) consistently achieves the highest SNR, *Fit*, and *Corr*, showcasing its superior ability to enhance signal quality across different noise conditions. These observations align with the observations from Fig. 4.

Power spectral density (PSD) analysis: The PSD provides a visual representation of how the power of a signal is distributed across different frequencies. The PSD was computed using MATLAB™'s `pwelch` function with a 'Hamming' window of 64 samples, an overlap of 32 samples, and 64 fast Fourier transform (FFT) points. Fig. 6 shows the PSD curves (for the same signal depicted in the qualitative results) for the clean, noisy, and signals denoised by various methods. It indicates that the PSD of the denoised signal obtained using the proposed DWT-UNet method (green) closely aligns with the PSD of the original signal (blue), indicating that the model effectively reduces noise while preserving the frequency characteristics of the original signal. Although the WaveCRN method achieves better noise reduction among the compared methods, it is challenging in real-life noise cases. The STFT-UNet, DWT-SURE, DCCRN, and DWT-ANN methods fail to preserve frequency contents. Notably, within the critical 0–400 Hz range, where respiratory sounds dominate, the proposed method effectively retains relevant information, unlike existing methods, which significantly alter the frequency characteristics.

Table III presents the mean absolute error (MAE) between the PSD of the clean signal and the denoised signal across different noise scenarios at -5 dB SNR in the frequency ranges of 0–1 kHz and 0–400 Hz, averaged across the test dataset. The table shows that the proposed method achieves the most substantial reductions in error values, showcasing its superior effectiveness in mitigating noise across all scenarios. Although WaveCRN produces better noise reduction than other existing methods, it cannot preserve low frequencies and is ineffective in noise reduction in real-life noise scenarios, as shown in Fig. 4. In contrast, the STFT-UNet model exhibits a greater loss of information, leading to a higher MAE, followed by DWT-ANN, DCCRN, and DWT-SURE. Additionally, the proposed method achieves the lowest MAE in the 0–400 Hz frequency range, where respiratory sounds are predominantly concentrated.

Statistical analysis: Since deep learning involves probabilities, statistical tests are often used to evaluate results. This study uses a paired t-test to compare the proposed and existing methods. This approach helped assess how well the proposed method performs in denoising respiratory sound signals compared to existing methods. Here, the null hypothesis (H_0) shows no difference between the methods' performances. A value of $h = 1$ means the null hypothesis is rejected at the Alpha significance level, while $h = 0$ means it is not rejected. In all

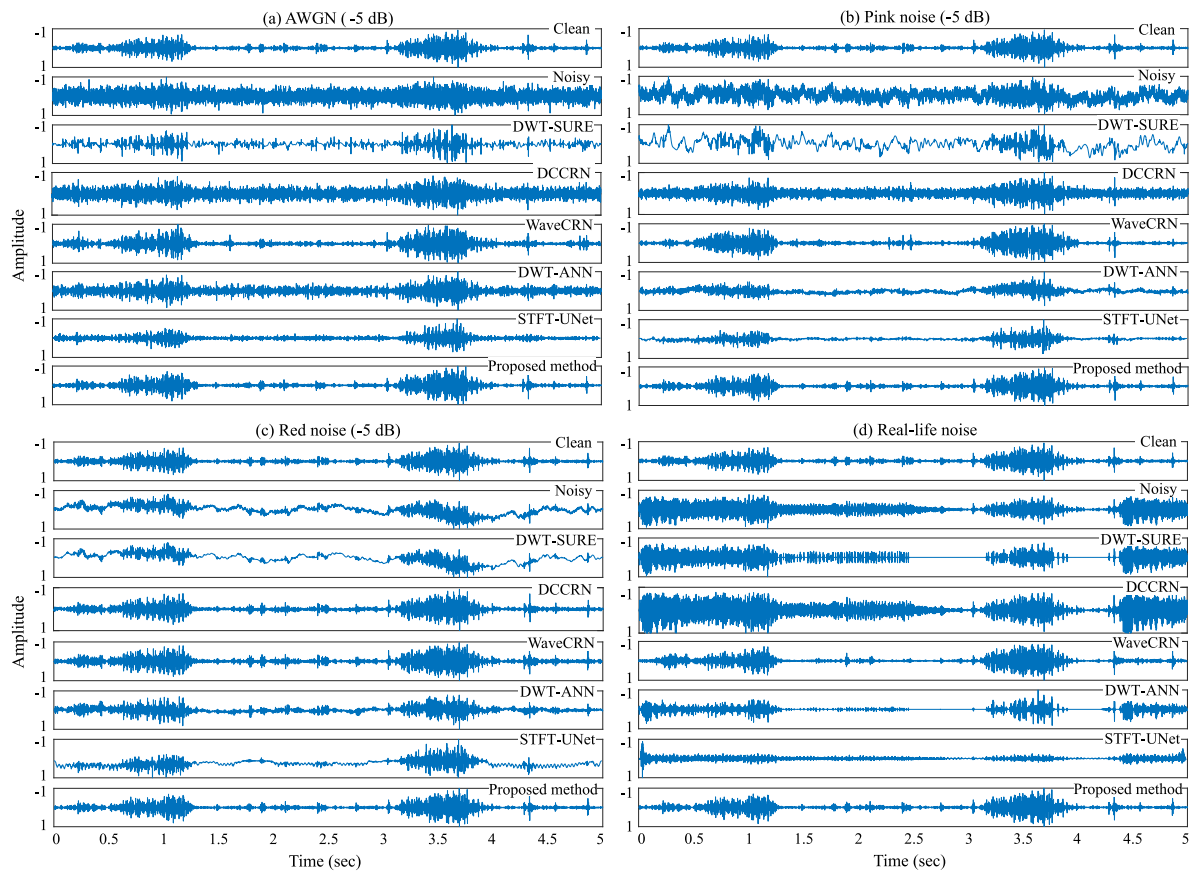


Fig. 4. Denoising results across various noise cases: (a) AWGN at -5 dB, (b) Pink noise at -5 dB, (c) Red noise at -5 dB, and (d) Real-life noise.

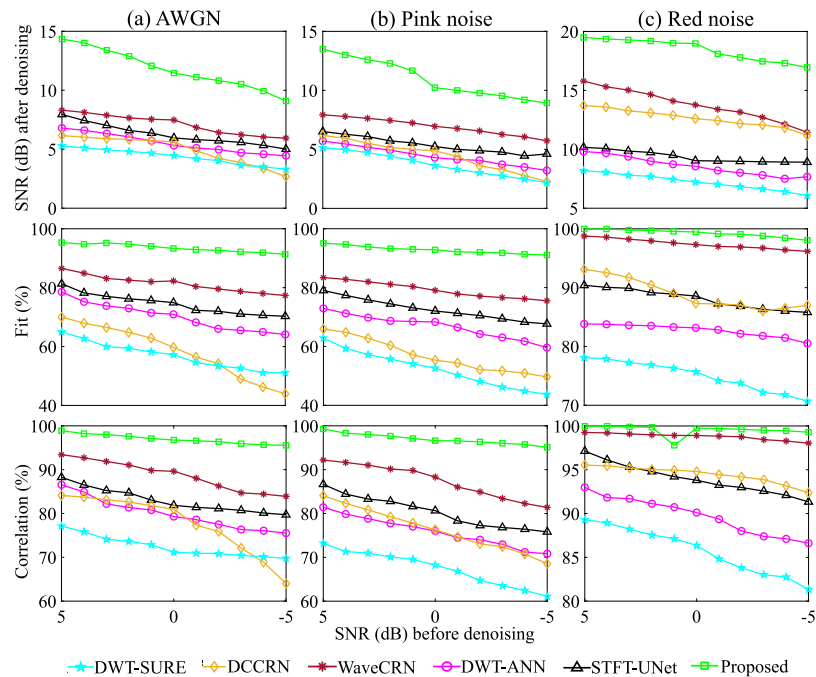
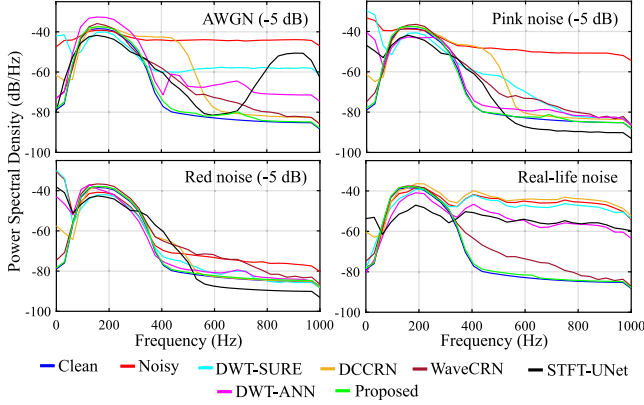


Fig. 5. Quantitative results SPRS-23 dataset at different SNR levels across various noise scenarios: (a) AWGN, (b) Pink, and (c) Red noise.

Table III

Average MAE between PSD of clean and denoised signal for 0-1 kHz and 0-400 Hz frequency ranges at SNR -5 dB.

	AWGN		Pink noise		Red noise		Real-life noise	
	0-1 kHz	0-400 Hz	0-1 kHz	0-400 Hz	0-1 kHz	0-400 Hz	0-1 kHz	0-400 Hz
Noisy	27.64	10.05	24.73	11.71	8.96	9.10	24.42	1.61
DTW (SURE) [13]	18.47	8.67	9.75	10.19	5.14	9.59	23.73	2.47
DCCRN [21]	9.41	4.88	7.40	4.48	3.32	4.96	26.58	5.36
WaveCRN [16]	6.33	3.28	5.91	3.36	5.47	2.60	4.84	1.52
DWT-ANN [9]	9.35	5.78	5.02	8.74	4.20	7.69	18.79	3.85
STFT-UNet [24]	8.13	4.62	6.48	7.51	7.58	7.06	21.54	12.11
Proposed method	0.99	0.64	1.16	1.02	0.56	0.41	0.72	0.66

**Fig. 6.** PSD curves of denoised signals using various methods across different noise scenarios.**Table IV**

Pair t-test for the proposed model with the compared methods.

Model	AWGN		Pink noise		Red noise	
	h	p	h	p	h	p
Proposed vs DWT-SURE [13]	1	0.0167	1	0.0059	1	0.0005
Proposed vs DCCRN [21]	1	0.0291	1	0.0062	1	0.0247
Proposed vs WaveCRN [16]	1	0.0360	1	0.0354	1	0.0132
Proposed vs DWT-ANN [9]	1	0.0184	1	0.0102	1	0.0012
Proposed vs STFT-UNet [24]	1	0.0156	1	0.0211	1	0.0038

scenarios shown in Table IV, the value of h is consistently 1, with p -values below the 0.05 significance level. This result leads to rejecting the null hypothesis, indicating a significant difference between the proposed and other methods. The analysis, with a 95% confidence interval, confirms the superior performance of the proposed method.

3.3.2. Results on ICBHI-17 dataset

Table V presents the quantitative performance on the ICBHI-17 dataset, averaged across the test dataset. It is evident from the table that the proposed method outperforms in terms of all metrics, especially under real-life noise conditions, where other denoising methods struggle to maintain signal quality. The proposed method improved the SNR of the signal from -5 dB to 8.17 dB (AWGN), 7.12 dB (pink noise), and 14.17 dB (red noise), demonstrating its robustness in restoring clean signals from noisy ones. The proposed method consistently produced Fit and $Corr$ above 90% across different noise conditions, except in real-life noise, where Fit is 88.16%, which is still 28% higher than the second-best result achieved by STFT-UNet. This significant improvement underscores the proposed method's effectiveness in retaining the respiratory information structure while minimizing noise, making it more suitable for real-world applications.

Fig. 7 further depicts the quantitative performance across three noise types evaluated over SNR ranging from 5 to -5 dB, similar to

the SPRS-23 dataset. The results highlight that the proposed method (green) exhibits a consistent superior performance across all noise conditions. For instance, in AWGN, the proposed method sustains an SNR close to 9 dB. Among the compared methods, STFT-UNet (black) produces higher performance metrics. The proposed method faces a gradual reduction in the Fit and $Corr$ with respect to a decrease in the SNR, in contrast to the compared methods for which metrics degrade significantly.

4. Discussion

The discussion covers the role of stride convolution in the proposed method's performance, the impact of denoising on sound event classification with and without noise, and the model's complexity.

4.1. Single-branch vs multi-branch UNet

The proposed method uses separate UNet models, multi-branch, for each level of DWT decomposition (D_2 , D_3 , D_4). The single-branch UNet inputs the concatenated or stacked DWT coefficients from multiple levels. The comparative results of both approaches, presented in Table VI, indicate that the multi-branch UNet (proposed method) produces significantly higher SNR than the single-branch UNet. The multi-branch UNet allows the network to learn frequency-specific noise and features effectively. At the same time, the single-branch UNet faces difficulty due to learning a large number of coefficients with mixed frequency characteristics. Although the number of parameters in the multi-branch UNet is three times higher than that of the single-branch UNet, the number of floating point operations (FLOPs) remains unchanged. Therefore, the overall computational complexity remains comparable.

4.2. Impact of stride convolution instead of maxpool

In the context of the UNet-based approach for denoising wavelet coefficients, a significant difference in information preservation was observed when adopting stride convolution instead of max-pooling [42]. Stride convolution reduces the dimensions of the input by skipping specific values. However, they keep more critical details by capturing key features because the convolutional filters efficiently capture essential features during down-sampling. Additionally, it preserves spatial relationships, which are necessary for recognizing patterns. In contrast, max-pooling discards valuable information by retaining only the maximum value within a window. For instance, under AWGN noise at -5 dB SNR, stride convolution achieved an SNR of 9.09 dB compared to 5.23 dB for max-pooling, with higher Fit (91.33% vs. 74.58%) and $Corr$ (95.54% vs. 85.32%). This indicates that stride convolution offers a more flexible approach than max pooling.

Table V
Quantitative results of denoising on ICBHI-17 dataset.

SNR before denoise	AWGN			Pink noise			Red noise			Real-life noise		
	-5 dB	0 dB		-5 dB	0 dB		-5 dB	0 dB		-		
Performance metrics	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)	SNR↑ (dB)	Fit↑ (%)	Corr↑ (%)
DWT-SURE [13]	2.05	40.32	57.63	3.19	46.93	66.52	1.11	35.59	52.14	2.62	42.45	63.98
DCCRN [21]	2.13	41.64	59.19	3.25	44.96	68.10	1.91	35.45	51.01	2.78	40.33	65.11
WaveCRN [16]	2.56	30.24	67.21	4.39	49.11	72.28	2.47	30.01	65.20	4.02	47.61	69.78
DWT-ANN [9]	3.12	45.36	60.32	4.25	52.12	64.00	2.35	40.19	52.64	3.03	44.33	63.65
STFT-UNet [24]	4.10	65.62	77.22	5.49	68.10	80.21	2.88	57.78	71.14	4.11	65.32	77.35
Proposed method	8.17	90.64	94.31	9.68	92.16	94.88	7.12	90.28	93.53	9.61	91.13	94.79
Improvement	+4.0	+25.0	+17.0	+4.1	+24.0	+14.6	+4.2	+32.5	+22.3	+5.5	+25.8	+17.4

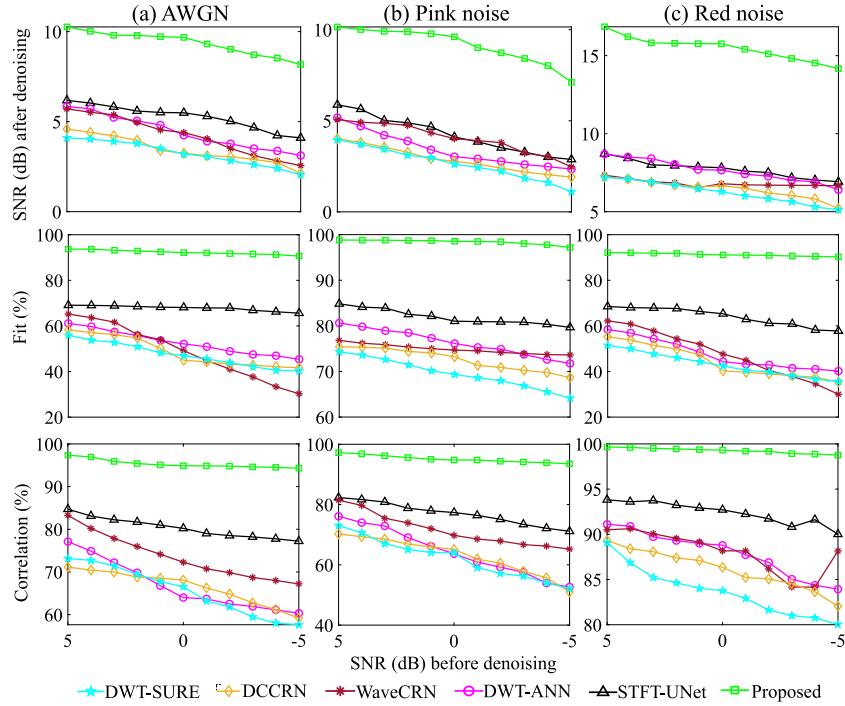


Fig. 7. Quantitative results of ICBHI-17 dataset at different SNR levels across various noise scenarios: (a) AWGN, (b) Pink, and (c) Red noise.

Table VI
Obtained SNR (dB) using single vs multi-branch UNet on SPRS-23 dataset.

SNR before denoise	AWGN		Pink noise		Red noise		Real-life noise
	-5 dB	0 dB	-5 dB	0 dB	-5 dB	0 dB	-
Single-branch UNet	4.96	6.86	4.61	6.05	9.91	10.60	4.91
Multi-branch UNet (proposed)	9.09	11.46	8.93	10.23	16.95	18.98	7.84

4.3. Impact of denoising on classification for noisy signals

The denoised signals are classified at the sound event level to demonstrate the impact of the denoising method. The SPRS-23 dataset contains 9780 sound events ranging from 137 ms to 7.15 s, including normal (7617), rhonchi (134), wheeze (646), stridor (47), coarse crackle (83), fine crackle (1221), and wheeze & crackle (32). Each event was segmented and padded to a uniform 5 s length. The classification method based on DT proposed in [32] is used to perform seven-class classification. A five-fold cross-validation approach is used to assess the classifier's performance.

Table VII presents the classification accuracy for the SPRS-23 dataset across various noise types at various SNR. For the SPRS-23 dataset, the DWT-SURE method consistently shows the lowest accuracy in all noise scenarios, followed by DCCRN, DWT-ANN, STFT-UNet, and WaveCRN. In contrast, the proposed method outperforms the others,

demonstrating the highest accuracy across all noise conditions. For instance, with AWGN noise at 5 dB, the proposed method achieves a classification accuracy of 73.85%, compared to the second-best result of 70.81% for WaveCRN, indicating a significant improvement. Moreover, for real-life noise, the proposed method achieves 72.64%, which is 3.45% higher than the second-best result.

4.4. Impact of denoising on classification for clean signals

Table VIII presents the classification results for the clean signals (without added noise). With the proposed denoising method, 79.89% accuracy (Acc) is achieved for the SPRS-23 dataset, a 2.33% increase over the classification without using any denoising method. It underscores the proposed method's effectiveness in suppressing the noise

Table VII

Classification accuracy (%) for the noisy signals on SPRS-23 dataset.

Denoise model	Classification model	AWGN			Pink noise			Red noise			Real-life noise
		-5 dB	0 dB	5 dB	-5 dB	0 dB	5 dB	-5 dB	0 dB	5 dB	
Before denoising	Decision tree [32]	22.03	24.51	28.20	21.23	22.19	25.62	30.96	37.84	43.25	40.81
DWT-SURE [13]		51.65	51.88	52.03	48.36	49.20	50.02	52.48	52.53	54.29	50.12
DCCRN [21]		50.31	51.26	54.99	50.35	51.03	53.26	62.95	64.87	67.70	43.52
WaveCRN [16]		69.86	70.04	70.81	68.45	68.85	69.20	72.60	72.98	73.17	69.19
DWT-ANN [9]		59.71	59.95	60.23	56.72	57.18	58.29	61.04	62.28	63.76	55.43
STFT-UNet [24]		64.13	65.28	65.40	61.94	62.05	63.39	65.91	66.40	67.17	60.48
Proposed method		71.96	72.05	73.85	72.48	73.96	73.88	74.84	75.73	76.04	72.64
Improvement		+2.10	+2.01	+3.04	+4.03	+5.11	+4.68	+2.24	+2.75	+2.87	+3.45

Table VIII

Classification results (%) for the clean signal on SPRS-23 dataset.

Denoise model	Classification model	Acc	Pre	Rec	F1
-	Decision Tree [32]	77.56	75.23	76.90	76.02
DWT-SURE [13]		56.21	53.52	57.90	55.60
DCCRN [21]		58.41	52.80	53.25	53.02
WaveCRN [16]		73.94	71.48	73.72	72.52
DWT-ANN [9]		65.40	63.11	64.53	63.77
STFT-UNet [24]		69.75	66.23	69.01	67.58
Proposed method		79.89	80.33	79.16	79.74

while preserving the signal information. On the other hand, the second-best result using the WaveCRN produced 73.94% accuracy, significantly lower than the proposed method. This pattern is also reflected in other performance metrics, including precision (Pre), recall (Rec), and F1-score (F1). In contrast, the compared denoising methods reduced the accuracy drastically. This indicates that the proposed method efficiently enhances signal quality, even for the recordings captured in a clinical environment with minimal contamination, whereas the compared methods affect the signal information in the noise suppression process.

4.5. Time and space complexity

Table IX presents the time complexity in terms of the number of FLOPs, space complexity as the number of learnable parameters, and test time per sample. While the proposed model has a higher space complexity, its FLOPs are comparable to other methods, an essential factor for implementation and execution. The testing time per sample for the proposed model is 50.36 ms, which is lower than STFT-UNet. Although DWT-ANN, DCCRN, and WaveCRN have lower complexity, they exhibited lower and inconsistent performance. DWT-SURE is a non-deep learning method that also struggles to suppress the noise. The proposed model provides effective denoising and significantly improves classification accuracy with intermediate time and space complexity. To make it more suitable for portable devices, TensorFlow Lite (TF-Lite) was used to reduce the model's size to 52.10 MB.

5. Conclusion

This work presents a novel approach for adaptively thresholding DWT coefficients using a UNet to suppress noise in respiratory sound signals, thereby eliminating the need for manual threshold parameter selection in traditional DWT-based denoising methods. Experimental results highlight the method's qualitative and quantitative superiority across performance metrics. Key findings demonstrate its effectiveness in reducing out-of-band and in-band noise while preserving essential low-frequency respiratory components. The analysis of its impact on

Table IX

Time and space complexity of models.

Model	FLOPs	Parameters (million)	Test time per sample (ms)
DCCRN [21]	2.55×10^{10}	3.67	24.03
WaveCRN [16]	5.57×10^8	4.65	10.20
DWT-ANN [9]	9.10×10^5	0.46	32.41
STFT-UNet [24]	9.41×10^9	2.47	88.64
Proposed method	D_2	1.15×10^9	50.36
	D_3	5.81×10^8	
	D_4	3.01×10^8	

sound event classification reveals a significant improvement in accuracy, even for clean signals, emphasizing the critical role of robust noise reduction in automatic respiratory health diagnosis under real-world conditions. The proposed method's strong performance is attributed to leveraging the multi-resolution capabilities of DWT and the multi-scale feature extraction of UNet. Replacing max-pooling with stride convolution further enhances UNet's ability to handle complex noise patterns. This model not only provides effective denoising but also significantly boosts classification accuracy while maintaining intermediate time and space complexity, making it well-suited for deployment on portable devices. Additionally, the approach holds potential for adaptation to other signal-processing tasks across various domains for noise suppression. Future work will focus on integrating attention mechanisms to refine the model's ability to target specific frequencies and channels, further enhancing its performance.

CRediT authorship contribution statement

Subhasikta Behera: Software, Data curation, Writing – original draft, Methodology, Writing – review & editing, Resources, Investigation, Visualization, Conceptualization, Validation, Formal analysis. **Puneet Kumar Jain:** Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization, Funding acquisition, Data curation, Supervision, Project administration, Formal analysis, Writing – review & editing, Visualization, Resources. **Sambit Bakshi:** Investigation, Conceptualization, Supervision, Writing – review & editing, Funding acquisition, Validation, Resources, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Puneet Kumar Jain reports financial support was provided by Science and Engineering Research Board. Sambit Bakshi reports financial support was provided by Vishlesan I-Hub Foundation, IIT Patna (NMCPs-DST), Government of India. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my code and public dataset used for the study in the manuscript.

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